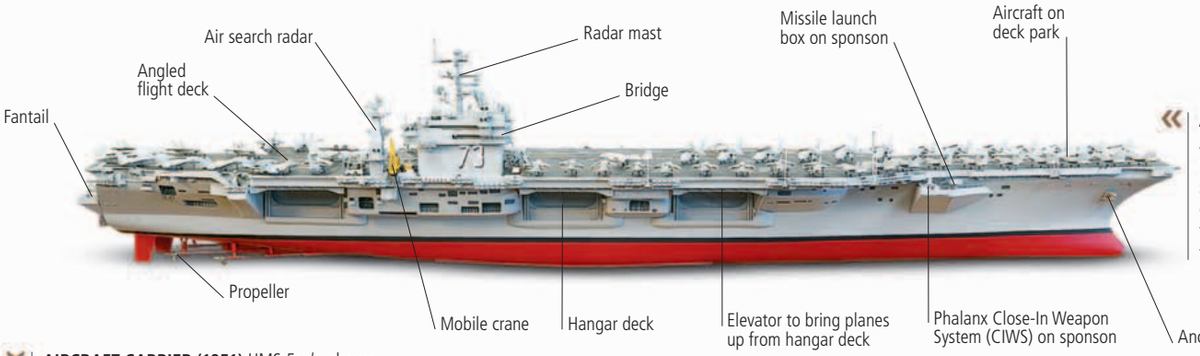
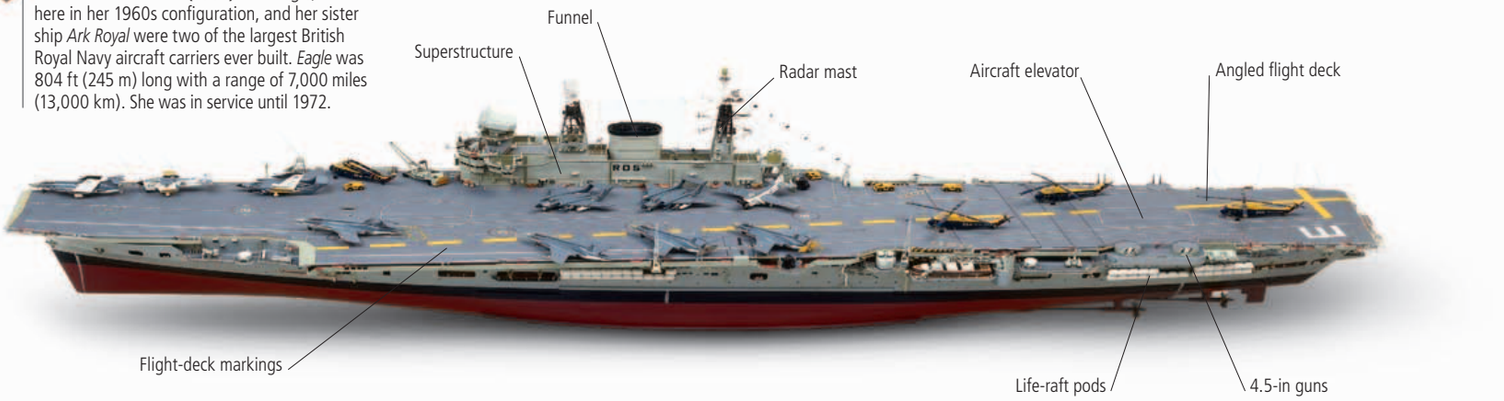
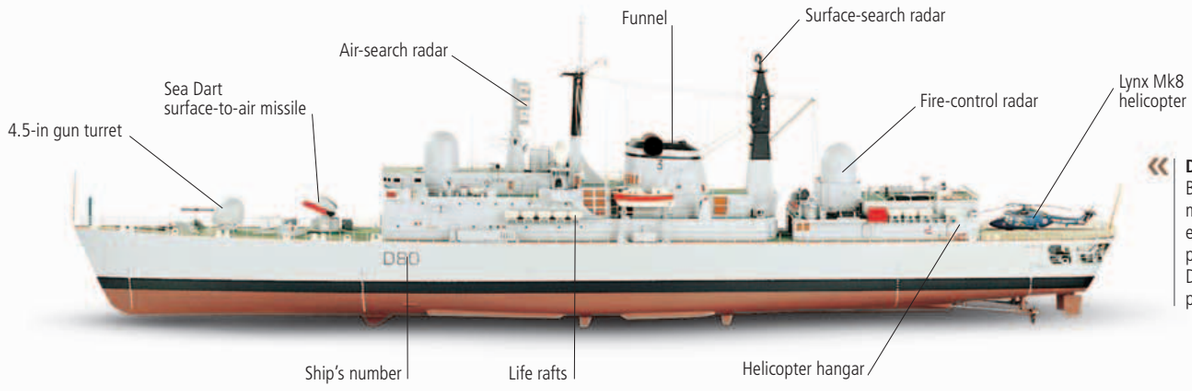
MODERN WARSHIPS



AIRCRAFT CARRIER (1990) Aircraft carriers such as the USS *George Washington* are powered by nuclear fuel—she has two on-board reactors and can operate for up to 18 years before having to refuel. She carries 85 aircraft, including fighter, strike, and transport planes.

AIRCRAFT CARRIER (1951) HMS *Eagle*, shown here in her 1960s configuration, and her sister ship *Ark Royal* were two of the largest British Royal Navy aircraft carriers ever built. *Eagle* was 804 ft (245 m) long with a range of 7,000 miles (13,000 km). She was in service until 1972.





DESTROYER (1971) By the 1970s destroyers had evolved into missile-armed air defense ships for fleet escort. The Royal Navy's Type 42 class (left) performed this function in the Falklands War. Destroyers have since become larger multi-purpose missile platforms.

SINCE THE END of World War II, naval warfare has been dominated by the aircraft carrier, the submarine, and the guided missile. The age of the battleship was over, even if big guns still served a purpose for shore bombardment. Aircraft carriers began to grow in size so that they could operate jet aircraft. By the 1970s, however, only the United States could afford to operate a fleet of full-size carriers—these were virtual floating cities, with enough air power to sustain an independent air campaign and were driven by nuclear powerplants that allowed them to stay at sea for an almost indefinite period of time. Smaller carriers were still deployed by some of the less affluent countries, and by the US Navy as "assault ships," operating helicopters and vertical/short take-off and landing (V/STOL) jet aircraft.

Nuclear-powered submarines, capable of phenomenal submerged endurance and speed, were equipped with intercontinental ballistic missiles as part of the nuclear armory of major powers. Hunter-killer submarines were also built, chiefly designed to hunt down enemy submarines. The days of pitched battles between surface fleets seemed over, replaced by a game of underwater cat-and-mouse between nuclear-armed or powered submarines.

GUIDED MISSILES

The development of guided missiles dramatically altered the nature of surface warfare at sea. Guns were reduced to a subsidiary role, with most surface vessels retaining only a single automatic cannon on the foredeck. By this point in time warships effectively became floating electronics and missile platforms. The destroyer ceased to have a role as a torpedo-attack ship, assuming a range of roles including fleet defense against air attack using anti-aircraft missiles. Frigates tended to specialize in anti-submarine warfare. Both vessels were driven by gas-turbine engines and were designed to be able to carry at least one helicopter. The offensive role of naval guns, meanwhile, was largely taken over by ship-killing missiles—which could be carried even by quite small patrol boats—and by Cruise missiles that were used for the bombardment of land targets. To defend against attack, warships began to deploy a range of electronic countermeasures designed to confuse missile guidance systems. Control rooms on modern naval vessels became vastly complex suites of electronic equipment, including links to surveillance satellites.